



Operating Instructions

Diesel engine
12V4000P11
16V4000P11

MS15004/04E

Engine model	kW/cyl.	Application group
12V4000P11	110 kW/cyl.	4A, continuous operation, unrestricted
16V4000P11	110 kW/cyl.	4A, continuous operation, unrestricted

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to §15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.

 **WARNING:** Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

Table of Contents

1 Preface		5 Operation	
1.1 Preface	7	5.1 Preparations for putting into operation after extended out-of-service periods (>3 months)	66
2 Safety		5.2 Preparations for startup after scheduled out-of-service-period	67
2.1 Important provisions for all products	8	5.3 Starting the engine in manual mode (test run)	68
2.2 Intended use of all products	9	5.4 Re-starting the engine following an automatic safety shutdown	69
2.3 Personnel and organizational requirements	10	5.5 Operational monitoring	70
2.4 Safety regulations for commissioning and operation	11	5.6 Engine – Stopping in manual mode (test run)	71
2.5 Safety regulations for assembly, maintenance, and repair work	13	5.7 Emergency engine stop	73
2.6 Fire and environmental protection, indirect materials	17	5.8 After stopping the engine – Putting the engine out of operation	74
2.7 Standards for warning messages in the text and highlighted information	20		
2.8 ATEX (where applicable)	21	6 Maintenance	
3 Transport		6.1 Maintenance task reference table [QL1]	75
3.1 Transport	22	7 Troubleshooting	
3.2 Lifting specifications	24	7.1 Troubleshooting	76
3.3 Transport locking device for crankshaft	25	7.2 Fault display on printed circuit board SAM	79
4 Product Summary		7.3 Fault messages on Engine Control Unit ECU 4	88
4.1 Plant Overview	28	8 Task Description	
4.1.1 Purpose	28	8.1 Engine	99
4.1.2 ECS-5 – Purpose of the units	35	8.1.1 Cranking engine manually	99
4.1.3 MDEC – Functions	38	8.1.2 Engine – Turning on starting system	101
4.1.4 Peripheral Interface Modules (PIMs)	43	8.1.3 Engine – Test run	102
4.1.5 Printed circuit board – SCB 3	45	8.2 Cylinder Liner	103
4.1.6 LOP – Control elements	48	8.2.1 Cylinder liner – Endoscopic examination	103
4.1.7 Data acquisition and data transmission	50	8.2.2 Cylinder liner – Instructions and comments on endoscopic and visual examination	105
4.2 Engine Layout	51	8.3 Crankcase Breather	107
4.2.1 Engine – Overview	51	8.3.1 Crankcase breather – Oil separator element replacement	107
4.2.2 Sensors and actuators – Overview	52	8.4 Valve Drive	109
4.3 Engine Side and Cylinder Designations	57	8.4.1 Valve gear – Lubrication	109
4.3.1 Engine side and cylinder designations	57	8.4.2 Valve clearance – Check and adjustment	110
4.4 Firing Sequence	58	8.4.3 Cylinder head cover – Removal and installation	112
4.4.1 Firing order	58	8.5 Injection Pump / HP Pump	113
4.5 Technical Data	59	8.5.1 HP fuel pump – Relief bore check	113
4.5.1 Engine – Main dimensions	59		
4.5.2 ENGINE DATA 12/16V4000P11	60		
4.5.3 SAM – Technical data	63		
4.5.4 SCB 3 – Technical data	65		

8.6	Injection Valve / Injector	114	8.15	Low-Temperature Circuit	157
8.6.1	Injector – Replacement	114	8.15.1	Charge-air coolant – Filling	157
8.6.2	Injector – Removal and installation	115	8.15.2	Charge-air coolant – Draining	160
8.7	Battery-Charging Generator	120	8.15.3	Charge-air coolant – Change	162
8.7.1	Battery-charging generator – Cleaning	120	8.15.4	Charge-air coolant level – Check	163
8.7.2	Battery-charging generator – Check	124	8.15.5	Charge-air coolant pump – Relief bore check	164
8.8	Fuel Filter	125	8.16	Engine Mounting / Support	165
8.8.1	Fuel filter – Replacement	125	8.16.1	Engine mounting – Checking resilient element	165
8.8.2	Fuel prefilter – Differential pressure gauge check and adjustment	127	8.17	Wiring (General) for Engine/Gearbox/Unit	166
8.8.3	Fuel prefilter – Draining	128	8.17.1	Engine cabling – Check	166
8.8.4	Fuel prefilter – Flushing	130	8.18	Accessories for (Electronic) Engine Governor / Control System	167
8.8.5	Fuel prefilter – Filter element replacement	133	8.18.1	Engine Monitoring Unit and connectors – Cleaning	167
8.9	Charge-Air Cooling General, Left-Hand Side	135	8.18.2	Engine Monitoring Unit – Plug connection check	168
8.9.1	Intercooler – Checking condensate drain for coolant discharge and obstruction	135	8.18.3	Engine governor – Removal and installation	169
8.10	Air Filter	136	8.18.4	Engine governor and connectors – Cleaning	170
8.10.1	Air filter – Replacement	136	8.18.5	Engine governor plug connections – Check	171
8.10.2	Air filter – Removal and installation	137	9	Engine Governor Assignment and Circuitry	
8.11	Air Intake	138	9.1	Engine Governor	172
8.11.1	Contamination indicator – Signal ring position check	138	9.1.1	Circuitry of Engine Control Unit channels	172
8.11.2	Emergency air shutoff flaps – Lubrication	139	9.1.2	I/O assignment of MDEC control unit	181
8.11.3	Emergency air-shutoff flaps – Functional check with electric actuation	140	9.2	Engine Monitoring	183
8.12	Lube Oil System, Lube Oil Circuit	141	9.2.1	LOP and connectors – Cleaning	183
8.12.1	Engine oil – Change	141	9.2.2	LOP – Visual inspection	184
8.12.2	Engine oil level – Check	143	9.2.3	LOP – Running test procedures	186
8.12.3	Engine oil – Sample extraction and analysis	144	9.3	Connector Assignment	188
8.13	Oil Filtration / Cooling	145	9.3.1	Connector pin assignment	188
8.13.1	Engine oil filter – Replacement	145	10	Appendix A	
8.13.2	Centrifugal oil filter – Cleaning and filter sleeve replacement	147	10.1	Abbreviations	197
8.14	Coolant Circuit, General, High-Temperature Circuit	149	10.2	Contact person / Service partner	201
8.14.1	Engine coolant – Filling	149	11	Appendix B	
8.14.2	Engine coolant – Draining	151	11.1	Standard Tools and Special Tools	202
8.14.3	Engine coolant – Change	152	11.2	Index	208
8.14.4	Engine coolant level – Check	153			
8.14.5	Engine coolant pump – Relief bore check	154			
8.14.6	Engine coolant – Sample extraction and analysis	155			

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:<http://www.mtu-solutions.com>

2.2 Intended use of all products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.